Literature Synthesis

Background Physics for the Radioisotope VUV Scintillation Light Source Paper

Synthesis document — not for direct inclusion in paper

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Contents

1 Overview and Narrative Logic

The five sections below provide the physics needed to understand a xenon second-continuum source with electroluminescent (EL) amplification. The logical story is:

1. Rare gases scintillate via excimer emission ($\mathrm{R}_2^* \to 2\mathrm{R} + h\nu$).
2. At operating pressures ($\gtrsim 400$ mbar), the emission is the *second* continuum—a narrow VUV band.
3. Ionizing radiation creates both excitation and ionization; LET affects the relative contributions but in the gas phase the total yield is well-approximated by W-values.
4. Electroluminescence provides sub-avalanche amplification that scales linearly with E/P, localizing photon production near the high-field electrode.
5. Therefore: a sealed radioisotope + EL electrode is a compact, spectrally pure excimer line source.

Keep the paper's Background section to 4–6 paragraphs. It should convey mechanism, scaling laws, and key numbers—not survey all rare-gas UV literature.

2 Rare-Gas Excimer Formation

When a rare-gas atom R is excited, it cannot form a stable ground-state molecule because the rare-gas dimer ground-state potential is purely repulsive. The *excited* state, however, is bound: the excimer R_2^* has a well-defined potential minimum. Upon radiative decay, the system drops onto the repulsive ground-state curve and immediately dissociates, $\mathrm{R}_2^* \to 2\mathrm{R} + h\nu$, emitting a VUV photon. Because the ground state is repulsive, there is no reabsorption at the emission wavelength—rare-gas excimers are intrinsically transparent to their own luminescence [1].

2.1 Two Pathways to R_2^*

Ionizing radiation deposits energy through both excitation and ionization. Both channels feed the same R_2^* excimer state [2, 3]:

Excitation channel.

$$
\begin{aligned}
e^- + \mathrm{R} &\to \mathrm{R}^* + e^- && \text{(electronic excitation)} && (1)\\
\mathrm{R}^* + 2\mathrm{R} &\to \mathrm{R}_2^{**} + \mathrm{R} && \text{(three-body, vibrationally excited)} && (2)\\
\mathrm{R}_2^{**} + \mathrm{R} &\to \mathrm{R}_2^* + \mathrm{R} && \text{(vibrational relaxation)} && (3)\\
\mathrm{R}_2^* &\to 2\mathrm{R} + h\nu && \text{(VUV photon, second continuum)} && (4)
\end{aligned}
$$

Ionization + recombination channel.

$$e^- + \mathrm{R} \to \mathrm{R}^+ + 2e^- \quad \text{(ionization)} \tag{5}$$

$$\mathrm{R}^+ + 2\mathrm{R} \to \mathrm{R}_2^+ + \mathrm{R} \quad \text{(ion clustering)} \tag{6}$$

$$\mathrm{R}_2^+ + e^- \to \mathrm{R}^{**} + \mathrm{R} \quad \text{(dissociative recombination} \to \text{Rydberg atom)} \tag{7}$$

$$\mathrm{R}^{**} + 2\mathrm{R} \to \mathrm{R}_2^{**} + \mathrm{R} \quad \text{(three-body, from Rydberg state)} \tag{8}$$

$$\mathrm{R}_2^{**} + (\mathrm{R}) \to \mathrm{R}_2^{*} + (\mathrm{R}) \quad \text{(predissociation to S levels)} \tag{9}$$

$$\mathrm{R}_2^{*} \to 2\mathrm{R} + h\nu \quad \text{(VUV photon)} \tag{10}$$

The dissociative recombination step (Eq. 7) does not produce R_2^* directly. Instead, as shown explicitly by Lorents [4] (Eqs. 4–7 of that work), the dimer ion R_2^+ captures an electron into an atomic Rydberg state R^{**} (the p-levels, corresponding to the 4p or 6p configurations of Ar and Xe respectively). This Rydberg atom then undergoes three-body association (Eq. 8) to form the vibrationally excited excimer R_2^{**}, which predissociates to the S levels that populate the radiating excimer R_2^* [4]. The "relaxation" in the ionization channel is therefore electronic (a Rydberg cascade from p-levels down to S levels), not vibrational. This is why the ionization channel does not require an explicit vibrational-relaxation step analogous to Eq. 3 in the excitation channel: the pathway through intermediate atomic Rydberg states bypasses the vibrationally unrelaxed R_2^{**} regime that produces the first continuum in the excitation channel.

Both pathways ultimately produce the same R_2^* state and the same emission spectrum [2]. Which pathway dominates depends on the applied electric field (which separates electrons from ions, suppressing recombination), gas pressure, and primary-particle linear energy transfer (LET). Saito et al. [5] confirmed the two-channel picture through simultaneous absolute measurements of ionization electrons and VUV photons from 5.49 MeV α particles in rare gases: as the applied field is increased, the charge yield rises and the scintillation yield falls, demonstrating the anti-correlation between the two channels.

The ion-clustering step (Eq. 6) and the dissociative recombination (Eq. 7) involve ion–molecule and electron–ion reactions whose rates were studied theoretically by Flannery [6, 7]. Rate constants for the de-excitation of rare-gas metastable atoms entering the excitation channel (Eq. 2) were measured by Velazco, Kolts & Setser [8] using a flowing afterglow technique for $\mathrm{Ar}(^3P_2)$, $\mathrm{Ar}(^3P_0)$, $\mathrm{Kr}(^3P_2)$, and $\mathrm{Xe}(^3P_2)$ at 300 K.

2.2 Excimer States and Emission Wavelengths

The R_2^* excimer supports two emitting states: the singlet $^1\Sigma_u^+$ (fast component, nanosecond lifetime) and the triplet $^3\Sigma_u^+$ (slow component, much longer lifetime). Both decay to the ground $^1\Sigma_g^+$ curve and contribute to the second continuum. The triplet radiative lifetimes differ dramatically across the rare-gas family [9]:

Gas	$^3\Sigma_u^+$ lifetime
Ar	~ 3200 ns
Kr	~ 1700 ns
Xe	~ 90 ns

Xenon's short triplet lifetime makes it the preferred choice for moderate to high event-rate applications.

The second-continuum peak wavelengths and bandwidths for the three heavy rare gases, as measured in gas proportional scintillation counter (GPSC) experiments, are [9, 10]:

Gas	Peak wavelength	FWHM
Ar	128 nm	~ 10 nm
Kr	147–148 nm	~ 12 nm
Xe	171–173 nm	~ 10–14 nm

These values agree closely between primary (passive) scintillation, electrical discharge measurements (Tanaka et al., cited in [9]), and secondary (EL) scintillation, confirming that the same R_2^* mechanism is responsible in all cases [9].

3 First and Second Continua: Pressure Dependence

The emission spectrum of an excited rare gas changes qualitatively with pressure because pressure controls the competition between radiative decay of isolated excited atoms and the collisional formation and vibrational relaxation of excimers [11, 10]. Three regimes are distinguishable:

Low pressure ($\lesssim 10$ mbar for Xe). At very low pressure, excited atoms decay radiatively before undergoing significant three-body collisions. In xenon, the allowed transition $\mathrm{Xe}^*(^1P_1) \to \mathrm{Xe} + h\nu$ (130 nm) and the intercombination transition $\mathrm{Xe}^*(^3P_1) \to \mathrm{Xe} + h\nu$ (147 nm) produce atomic resonance lines [10]. No excimer emission occurs.

Intermediate pressure: first continuum ($\sim$10–400 mbar for Xe). As pressure rises, three-body collisions (Eq. 2) begin converting R^* into vibrationally excited excimers R_2^{**} before radiative decay. These nascent dimers can emit *before* full vibrational relaxation, producing the **first continuum**: a broad, short-wavelength band from the $(\mathrm{Xe}_2^*)^{v\neq 0} \to 2\mathrm{Xe} + h\nu$ bound-free transition [10]. For xenon, the first continuum is centered near 147 nm. The transition from isolated-atom emission to excimer emission shifts the photon energy to lower values (longer wavelengths) as pressure increases [11].

High pressure: second continuum ($\gtrsim 400$ mbar for Xe). Above ~ 400 mbar, vibrational relaxation (Eq. 3) is rapid compared to the radiative lifetime of R_2^{**}. The population is efficiently transferred to the $v = 0$ level of R_2^* before emission occurs. The resulting **second continuum** is the $\mathrm{R}_2^*(v = 0) \to 2\mathrm{R} + h\nu$ transition: a narrow band centered at 172 nm with FWHM ~ 10 nm for xenon [11, 9]. Takahashi et al. [10] confirmed this behavior directly by measuring the proportional scintillation spectra of Ar, Kr, and Xe at 760 Torr in a GPSC: in each case only the respective second continuum was observed. Monteiro et al. [11] state explicitly: *"Above 400 mbar, the second continuum is dominant and the secondary scintillation presents only a narrow peak, ~ 10 nm FWHM, centred at 172 nm."*

3.1 Practical implication

Operating at 3 bar places the device firmly in the second-continuum regime. The emission from both passive scintillation and electroluminescence is the same narrow 172 nm band from Xe_2^*. The energy loss due to vibrational relaxation—the difference between the first and second continuum photon energies—is small (a few hundred meV) and is absorbed in collisions; Monteiro et al. [11] calculate that at $E/P = 4\ \mathrm{kV\,cm^{-1}\,bar^{-1}}$ the excitation efficiency (fraction of electron energy going into electronic excitation) reaches $\sim 95\%$, with vibrational relaxation losses being only a few percent.

4 Ionizing Radiation, LET, and Recombination

4.1 Track Structure and LET

The linear energy transfer (LET, or stopping power $-dE/dx$) of the primary particle determines the spatial density of excitation and ionization along the track. Alpha particles at a few MeV have high LET (∼MeV/cm at 3 bar Xe), creating dense, narrow ionization columns. Electrons from beta emitters or gamma-ray Compton scattering have much lower LET, depositing energy over longer, branched tracks [12].

In the gas phase at multi-bar pressures, the lower atomic density compared to liquid substantially reduces bimolecular reaction rates. Key consequences are:

1. **Columnar recombination** within the alpha track can occur at zero field, feeding the ionization+recombination excimer channel and thereby converting ionization into scintillation rather than into free charge. Suzuki, Ruan & Kubota [3] studied the time structure of this recombination luminescence in high-pressure Ar, Kr, and Xe excited by α particles. They observed two luminescence components: a fast component ($\lesssim 1\ \mu$s) from direct excitation and a slow component (microseconds) from recombination. The slow fraction increases with pressure and decreases with applied field, as expected.

2. **Bimolecular exciton quenching** ($\mathrm{R}^* + \mathrm{R}^* \to \mathrm{R} + \mathrm{R}^+ + e^-$), which is significant in liquid xenon at high excitation densities [12, 13], is much weaker in gas because the lower number density reduces the rate of these bimolecular collisions.

4.2 Recombination Suppression by Applied Field

Trindade et al. [2] report that electron–ion recombination in gaseous xenon at 800 Torr is negligible for drift fields above $\sim 0.05\ \mathrm{V\,cm^{-1}\,Torr^{-1}}$, even for the dense ionization tracks of α particles. At the drift and EL fields relevant to a working device (of order 10 V/cm/bar or higher), essentially all primary ionization electrons escape columnar recombination and are available for drift and EL amplification.

4.3 LET Effects on Scintillation Yield in Gas

In condensed-phase scintillators, the Birks quenching law describes how light yield per unit path length decreases at high dE/dx due to bimolecular quenching [12]. In the gas phase the situation is more nuanced. Several GPSC and gas-TPC studies find no strong evidence for α-particle quenching of primary scintillation in xenon gas at multi-bar pressures under applied fields. This is qualitatively consistent with the lower gas-phase number density and with the argument that recombination—which scales with LET—feeds the excimer channel rather than quenching it.

Implication for our source: In passive (zero-field) operation, the α track produces a mixed fast+slow scintillation signal including a recombination component. In EL operation, the drift field suppresses recombination, so the primary signal is dominated by the direct excitation channel. In both modes, the EL photons at the needle tip are produced by the same sub-avalanche mechanism and are independent of primary particle LET.

5 Scintillation Yield and the W-Value

5.1 Definitions

The efficiency of a noble-gas scintillator is quantified by two mean energies:

- W_i: mean energy deposited per ion pair (ionization W-value, eV/ion pair).
- W_s (or W_p): mean energy deposited per emitted VUV photon (scintillation W-value, eV/photon).

Both are empirical quantities that absorb all non-radiative loss channels (sub-threshold elastic scattering, molecular vibrational energy, quenching). Because $W_s > W_i$, not all deposited energy emerges as photons.

5.2 Ionization W-Value

For gaseous xenon, $W_i \approx 21.6$–22.4 eV depending on particle type and energy. The value $W_i = 21.61$ eV is widely cited from NIST [13]. Monteiro et al. [11] use $W_i = 22.4$ eV for 5.9 keV X-rays, giving $N_e = 5900/22.4 \approx 263$ primary electrons per X-ray interaction. For comparison: $W_i \approx 24$ eV for Kr and ≈ 26 eV for Ar.

5.3 Scintillation W-Value

For soft X-ray excitation in gaseous xenon at ~ 800 Torr (the benchmark condition used in most GPSC studies):

- Trindade et al. [2]: $W_s = 80 \pm 12$ eV (5.9 keV X-rays, 800 Torr Xe).
- Fernandes et al. [14]: $W_s = 72 \pm 6$ eV (5.9 keV X-rays, 800 Torr Xe).

The ratio $W_s/W_i \approx 3$–4 reflects the fraction of deposited energy lost to non-radiative channels. At higher pressure, the three-body excimer-formation step (Eq. 2) becomes more efficient (shorter mean free path between three-body events), so a larger fraction of excited atoms convert to excimers before being quenched, and W_s is expected to decrease.

For α-particle excitation, W_s differs from the X-ray value because the excitation/ionization ratio changes with LET. Saito et al. [5] performed the most direct absolute measurement: simultaneous counting of ionization electrons and VUV photons from 5.49 MeV ^{241}Am α particles in Ar, Kr, and Xe. Their results provide absolute W_s values for α excitation and directly demonstrate the anti-correlation between charge yield and light yield as a function of applied field—confirming that recombination transfers charge into photons and vice versa.

5.4 Photon Budget for the ^{148}Gd Source

As a rough estimate for our device, using $W_s \approx 80$ eV (conservative gas-phase value at 3 bar, scaled from 800 Torr measurements) and the ^{148}Gd α energy of 3.18 MeV:

$$N_\gamma^{\text{passive}} \approx \frac{3.18 \times 10^6 \text{ eV}}{80 \text{ eV}} \approx 4 \times 10^4 \text{ photons per decay.} \tag{11}$$

At a visible trigger rate of ~ 2200 decays/s (half of the 4.4 kBq activity, the half facing into the gas), the passive emission is of order $\sim 10^8$ photons/s into 4π sr before any optical losses. This is a loose upper bound; the actual value depends on pressure, applied field, and LET corrections.

5.5 Gas Purity

Electronegative impurities—primarily O_2 and H_2O—suppress scintillation through two channels: (a) electron attachment, removing primary electrons before they produce EL, and (b) VUV photon absorption, since the absorption cross-sections of O_2 and H_2O at 172 nm are very large. Even at sub-ppm levels these impurities can reduce the observed light yield substantially. This motivates the use of high-purity gas (N5 or better), outgassed materials, and getter-based purification [11].

6 Electroluminescence and E/P Scaling

6.1 Mechanism

Electroluminescence (EL), also called *proportional scintillation* or *secondary scintillation*, occurs when free electrons drift through a region of elevated, sub-avalanche electric field. Between inelastic collisions, electrons gain energy from the field; if E/P is above the excitation threshold but below the ionization threshold, each inelastic collision produces an excited atom R^* rather than an ion pair. The excited atom rapidly undergoes three-body excimer formation and emits a VUV photon via the same mechanism as primary scintillation:

$$e^- + \mathrm{R} \to \mathrm{R}^* + e^- \quad \text{(electron-impact excitation, no ionization)} \tag{12}$$

$$\mathrm{R}^* + 2\mathrm{R} \to \mathrm{R}_2^* + \mathrm{R} \tag{13}$$

$$\mathrm{R}_2^* \to 2\mathrm{R} + h\nu \quad \text{(172 nm for Xe)} \tag{14}$$

The EL spectrum is therefore identical to the primary scintillation spectrum. Suzuki & Kubota [9] confirmed this directly, measuring GPSC emission spectra for Ar, Kr, and Xe at 760 Torr and finding second-continuum peak positions and widths in agreement with discharge measurements—ruling out bremsstrahlung as the mechanism (as once proposed by Askar'yan) and confirming excimer emission.

6.2 Historical development

The GPSC was first demonstrated by Conde & Policarpo in 1967 [15] using argon at $\sim$ 1.4 bar. They observed secondary light pulses above a threshold anode voltage, with pulse amplitude growing steeply with field. A light-multiplication factor of ~ 140 was reached before Townsend avalanche set in, and an energy resolution of $\sim 2.3\%$ FWHM for alpha particles was achieved—superior to conventional proportional counters because EL adds no multiplicative noise. Subsequent work by Conde, Policarpo & Alves [16] extended the technique to xenon and xenon mixtures: pure Xe and Ar–5% Xe gave the best EL light output, with pulse amplitudes $\sim 100\times$ those of CsI(Tl) scintillators at optimal bias. Best energy resolution was 3.3% for Ar–5% Xe at 1.36 kV.

6.3 E/P Scaling Law

The number of EL photons produced per drifting electron per unit path length scales approximately linearly with the *reduced electric field* E/P (electric field divided by gas number density, or equivalently field divided by pressure) above a threshold. This scaling arises because the electron mean free path $\propto 1/P$, so the energy gained between collisions $\propto E/P$.

The standard parameterization [11, 2] is:

$$L = \left(A\frac{E}{p} - B \right) \times p \quad [\text{photons } e^{-1} \text{ cm}^{-1}], \tag{15}$$

where E/p is in $\text{V cm}^{-1}\,\text{Torr}^{-1}$, p in Torr, and A, B are empirical constants. For xenon, Monteiro et al. [11] obtained:

$$\begin{aligned} A &= 0.1389 \text{ photons } e^{-1} \text{ V}^{-1}, \\ B &= 0.1325 \text{ photons } e^{-1} \text{ cm}^{-1} \text{ Torr}^{-1}, \\ (E/p)_{\text{th}} &= B/A \approx 0.954 \text{ V cm}^{-1}\,\text{Torr}^{-1} \approx 1.26 \text{ kV cm}^{-1}\,\text{bar}^{-1}. \end{aligned} \tag{16}$$

Monteiro et al. [11] also express the yield as an amplification parameter of **140 photons per kV** applied across the scintillation region. At their benchmark field of 4.1 $\text{kV cm}^{-1}\,\text{bar}^{-1}$ (1.52 bar Xe), the reduced yield is $Y/p = 466$ $\text{photons}\,e^{-1}\,\text{cm}^{-1}\,\text{bar}^{-1}$.

Note on units: different references use different unit systems for E/P (V/cm/Torr vs. kV/cm/bar, with conversion factor 1 kV/cm/bar = 0.750 V/cm/Torr). Be careful when comparing values across papers.

6.4 EL Operating Window

The EL regime is bounded above and below in E/P:

- **Lower bound:** EL threshold $(E/P)_{\text{th}} \approx 1.26$ $\text{kV cm}^{-1}\,\text{bar}^{-1}$ for Xe. Below this, electrons never acquire enough energy to excite gas atoms.

- **Upper bound:** Townsend ionization threshold $\approx$ 10–15 kV cm^{-1} bar^{-1} for Xe. Above this, electron multiplication (avalanche) begins, producing ion feedback and gain fluctuations.

In the EL window, excitation efficiency (fraction of electron energy going into atomic excitation rather than elastic recoil) rises steeply with E/P and reaches $\sim 95\%$ at 4 kV cm^{-1} bar^{-1} [11]. This makes EL an efficient VUV-photon generator with essentially no statistical multiplicative noise from the amplification process itself, giving better energy resolution than charge-multiplication counters for the same gain.

6.5 Pressure Dependence of EL Yield: High-Pressure Measurements

The linear E/P scaling of Eq. 15 was measured near atmospheric pressure by Monteiro et al. [11] (1.52 bar). Freitas et al. [17] extended these measurements to 2–10 bar using the same LAAPD-based absolute method, directly relevant to our 3 bar operating condition. Their key results are:

- The linear Y/p vs. E/P relationship is preserved at all pressures from 2 to 10 bar.
- At 2 bar the parameterization is $Y/p = 151\,E/p - 131$ [photons e^{-1} cm^{-1} bar^{-1}, with E/p in kV cm^{-1} bar^{-1}], giving an EL threshold of ~ 0.8 kV cm^{-1} bar^{-1}.
- The scintillation amplification parameter (photons per drifting electron per kV) is nearly pressure-independent from 2 to 5 bar (Table 1 of Freitas et al.):

Pressure (bar)	Amplification (photons e^{-1} kV^{-1})
2	141 ± 6
3	141 ± 7
4	142 ± 5
5	151 ± 5
8	170 ± 10
10	162 ± 10

- At 3 bar (our operating condition), the amplification parameter is 141 ± 7 photons/kV—consistent with the 1.52 bar measurement of Monteiro et al. [11].
- Above 8 bar the parameter levels off and even decreases slightly, attributed to non-negligible electron attachment at high gas density.

This is the most directly applicable EL dataset for our device. The 3 bar result confirms that the Monteiro et al. parameterisation can be used at our operating pressure without significant correction. The slight pressure increase (at most $\sim 20\%$ from 2 to 8 bar) is negligible at our operating point.

6.6 EL in Structured Detectors (THGEM)

For completeness, Cortesi et al. [18] measured the secondary scintillation yield produced during gas avalanche in multi-layer Thick Gas Electron Multiplier (M-THGEM) structures operated in low-pressure CF_4 (20–100 Torr) and Ar/5%Xe (20–100 Torr). These measurements probe EL generated *inside avalanche holes* rather than in a pure drift region, and are therefore not directly comparable to our sub-avalanche EL regime. Key findings:

- EL yield (photons per avalanche electron): ~ 0.4 ph/el in CF_4 at 20 Torr; ~ 0.1 ph/el in Ar/5%Xe at 20 Torr.
- Yield decreases significantly with increasing pressure in both gases; the Ar/Xe mixture shows a steeper pressure dependence.
- In Ar/5%Xe, the emission is dominated by the xenon second continuum (~ 178 nm), consistent with energy transfer from Ar_2^* or Ar^* to Xe atoms.
- Applications targeted: low-pressure optical-readout TPCs for rare isotope beam experiments and directional dark matter detection.

This work is relevant as an example of EL-based optical readout in non-uniform high-field structures, demonstrating the generality of the excimer emission mechanism across different electrode geometries and gas mixtures.

6.7 Absolute Measurements and Literature Discrepancies

Monteiro et al. [11] note that published experimental EL yield values at room temperature differ significantly from each other and are a factor of ~ 2 lower than Monte Carlo and Boltzmann-transport predictions. The cause of this discrepancy was not fully resolved in 2007. Their own measurement—using a driftless GPSC at 1.52 bar Xe with a large-area APD—gave 140 photons/kV, in good agreement with Monte Carlo predictions and with cryogenic measurements taken in equilibrium with liquid xenon, suggesting their method has lower systematic uncertainties than previous room-temperature experiments. Trindade et al. [2] report the same Monteiro 2007 parameterization as the reference value for their work.

6.8 Relevance to the Needle-Electrode Design

In our device, the needle electrode creates a highly non-uniform field. The region near the tip has E/P in or above the EL regime, while the bulk of the gas is well below threshold. Primary ionization electrons produced by the α source drift toward the tip, enter the high-field region, and produce EL photons localized to the near-tip volume. This provides both gain ($\sim 100\times$ observed in the prototype) and spatial localization—the key properties for a compact, partially collimatable VUV point source.

The EL spectrum at the needle is identical to the passive scintillation (172 nm, ~ 10 nm FWHM), so the device retains its spectral character at all bias settings. Because EL photon production is a fixed function of E/P and path length, the EL gain is largely independent of the primary-particle LET; it amplifies whatever primary ionization charge is delivered to the high-field region, making it equally effective for α-, β-, or γ-driven primary excitation.

7 Key Quantitative Values at a Glance

Quantity	Value	Conditions	Reference
Xe_2^* peak (2nd continuum)	172 nm, FWHM $\sim$10 nm	Xe >400 mbar	[11, 9]
Kr_2^* peak (2nd continuum)	147–148 nm	Kr	[10]
Ar_2^* peak (2nd continuum)	128 nm	Ar	[9, 10]
${}^3\Sigma_u^+$ lifetime, Xe	$\sim$90 ns	gas	[9]
Pressure for 2nd-continuum dominance (Xe)	$\gtrsim$400 mbar	Xe	[11]
W_i (Xe)	21.6 eV	gas	[13, 2]
W_s (Xe, 5.9 keV X-ray)	80 ± 12 eV	800 Torr	[2]
W_s (Xe, 5.9 keV X-ray)	72 ± 6 eV	800 Torr	[14]
EL threshold $(E/P)_{\rm th}$	$\sim$1.26 kV/cm/bar	Xe	[11]
EL amplification parameter	140 photons/kV	Xe, 1.52 bar	[11]
EL reduced yield at 4.1 kV/cm/bar	466 ph/e^-/cm/bar	Xe, 1.52 bar	[11]
EL slope A	0.139 ph/e^-/V	Xe	[2]
EL offset B	0.1325 ph/e^-/cm/Torr	Xe	[2]
EL amplification at 3 bar	141 ± 7 photons/kV	Xe, 3 bar	[17]
EL amplification at 8 bar	170 ± 10 photons/kV	Xe, 8 bar	[17]
EL yield (M-THGEM, CF_4)	$\sim$0.4 ph/avalanche e^-	20 Torr	[18]
EL yield (M-THGEM, Ar/5%Xe)	$\sim$0.1 ph/avalanche e^-	20 Torr	[18]

8 Suggested Background Narrative for the Paper

The following is a suggested paragraph-level outline for the 4–6 paragraph Background section. It prioritises mechanism, key numbers, and scaling laws over historical survey.

Paragraph 1 — Excimer formation. Rare gases scintillate via excimer emission. When ionizing radiation deposits energy in a noble gas, excited atoms R^* and electron–ion pairs are produced. Three-body collisions at moderate pressure convert R^* into bound excimer states R_2^*; ionized atoms undergo clustering and dissociative recombination to reach the same state.

The excimer decays to the repulsive ground state—which has no bound levels to re-absorb the photon—emitting a VUV photon. *Key refs:* [1, 9, 5].

Paragraph 2 — Second continuum and pressure. At operating pressures ($\gtrsim 0.5$ bar), vibrational relaxation of the nascent excimer is fast, and emission is from the lowest vibrational level ($v = 0$) of R_2^*—the second continuum. For xenon, this is a ~ 10 nm FWHM band at 172 nm; for krypton, 148 nm; for argon, 128 nm. At lower pressures, unrelaxed excimers emit the broader, shorter-wavelength first continuum. At 3 bar (our operating condition) only the second continuum is present. *Key refs:* [11, 9, 10].

Paragraph 3 — Photon yield and W-values. The mean energy deposited per VUV photon emitted, W_s, characterises scintillation efficiency. For xenon gas at $\sim$1 bar, $W_s \approx$ 72–80 eV for soft X-ray excitation [2, 14], compared to an ionisation W-value of $W_i =$ 21.6 eV [13]; the ratio $W_s/W_i \approx 3$–4 reflects non-radiative losses. For the 3.18 MeV α decays of ^{148}Gd, this yields $\sim 4 \times 10^4$ passive scintillation photons per decay. Gas purity at the ppm level is essential: electronegative impurities (O_2, H_2O) quench electrons and absorb 172-nm photons. *Key refs:* [2, 14, 5, 19].

Paragraph 4 — LET and gas-phase recombination. Alpha particles produce dense ionisation columns (high LET); in the condensed phase, bimolecular quenching substantially reduces scintillation yield [12]. In the gas phase at multi-bar pressures and under even a small applied drift field ($\gtrsim 0.05$ V/cm/Torr), columnar recombination is suppressed and bimolecular quenching is negligible [2], so the gas-phase yield is less sensitive to LET than in liquid. In zero-field operation, a slow (μs) recombination luminescence component contributes to the total light yield alongside the fast excitation component [3].

Paragraph 5 — Electroluminescence and E/P scaling. When free electrons drift through a moderate electric field (E/P above threshold but below the Townsend limit), they excite gas atoms through inelastic collisions without ionising them. Each excited atom forms an excimer and emits a second-continuum photon—the same mechanism and spectrum as primary scintillation. The EL photon yield per drifting electron per unit path length scales linearly with E/P above an onset field of ~ 1 kV/cm/bar for xenon, with an amplification parameter of ~ 141 photons/kV that is essentially pressure-independent from 2 to 5 bar [17, 11]. At 3 bar (our operating pressure), the measured value is 141 ± 7 photons/kV [17]. EL introduces no statistical multiplicative noise and can be localised to a small high-field region, making it ideal for producing a bright, spectrally pure, spatially defined VUV source. *Key refs:* [17, 11, 2, 9, 15].

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